MDS221 — Standard Mock Paper II

120 Questions = 240 Marks · Single Best Answer · Lectures 1–12

A second completely fresh set at the same easy, past-paper level as the first Standard Mock Paper — single-step recall, no clinical vignettes, no multi-stage reasoning. This round leans on the full facial expression muscle tables, the pharyngeal arch nerve associations, and the core neuromuscular junction facts that the first paper only touched lightly.

Checks run on this set: cross-checked against all five earlier mock papers, including the first Standard Mock Paper, to avoid repeats; all 120 answers verified against the lecture slide that teaches them; correct answers distributed exactly evenly across A–E (24 each); all five options in every question matched for length so nothing leaks; automated structural check returned zero errors.

01 NECK: TRIANGLES, VESSELS & NERVES

LECTURES 1-2

1. The posterior triangle of the neck is bounded anteriorly by which muscle?

(A) Scalenus anterior
(B) Trapezius
(C) Digastric
(D) Sternocleidomastoid
(E) Omohyoid

2. Which triangle of the neck lies within the lower part of the posterior triangle and contains the third part of the subclavian artery?

(A) Submandibular triangle
(B) Occipital triangle
(C) Subclavian triangle
(D) Muscular triangle
(E) Carotid triangle

3. Which muscle divides the posterior triangle into an upper occipital triangle and a lower subclavian triangle?

(A) Trapezius
(B) Superior belly of omohyoid
(C) Inferior belly of omohyoid
(D) Scalenus medius
(E) Sternocleidomastoid

4. Which nerve descends on the anterior surface of scalenus anterior to supply the diaphragm?

(A) Accessory nerve
(B) Ansa cervicalis
(C) Hypoglossal nerve
(D) Vagus nerve
(E) Phrenic nerve

5. The submandibular gland is bounded by the two bellies of which muscle?

(A) Mylohyoid
(B) Stylohyoid
(C) Digastric
(D) Omohyoid
(E) Geniohyoid

6. Which triangle of the neck is bounded by the two anterior bellies of digastric and the hyoid bone?

(A) Muscular triangle
(B) Submandibular triangle
(C) Occipital triangle
(D) Carotid triangle
(E) Submental triangle

7. Which vein passes through the substance of the parotid gland before joining the external jugular system?

(A) Facial vein
(B) Lingual vein
(C) Superior thyroid vein
(D) Retromandibular vein
(E) Anterior jugular vein

8. Which nerve supplies the sternohyoid, sternothyroid, and omohyoid muscles?

(A) Accessory nerve
(B) Ansa cervicalis
(C) Phrenic nerve
(D) Hypoglossal nerve directly
(E) Vagus nerve

9. Which muscle is the exception among the infrahyoid muscles, receiving direct first cervical fibres via the hypoglossal nerve rather than the ansa cervicalis?

(A) Sternohyoid
(B) Thyrohyoid
(C) Omohyoid
(D) Geniohyoid
(E) Sternothyroid

10. Which nerve crosses superficial to the internal and external carotid arteries to reach the tongue?

(A) Ansa cervicalis
(B) Hypoglossal nerve
(C) Glossopharyngeal nerve
(D) Accessory nerve
(E) Vagus nerve

11. Which artery is the first branch of the external carotid artery?

(A) Superior thyroid artery
(B) Facial artery
(C) Maxillary artery
(D) Occipital artery
(E) Lingual artery

12. The internal jugular vein terminates by joining the subclavian vein to form which structure?

(A) Innominate artery
(B) Azygos vein
(C) Superior vena cava directly
(D) Brachiocephalic vein
(E) External jugular vein

13. Within the carotid sheath, which nerve runs the whole length of the neck alongside the great vessels?

- **(A)** Glossopharyngeal nerve
- **(B)** Hypoglossal nerve
- **(C)** Accessory nerve
- **(D)** Phrenic nerve
- **(E)** Vagus nerve

14. The thyroid gland is enclosed by which layer of deep cervical fascia?

- **(A)** Prevertebral fascia
- **(B)** Investing fascia
- **(C)** Pretracheal fascia
- **(D)** Alar fascia
- **(E)** Buccopharyngeal fascia

15. Omohyoid has two bellies joined by an intermediate tendon. Which belly lies closer to the hyoid bone?

- **(A)** Superior belly
- **(B)** Neither belly attaches near the hyoid
- **(C)** Both bellies are equidistant
- **(D)** Inferior belly
- **(E)** The intermediate tendon attaches to the hyoid instead

16. Which muscle has two bellies connected by an intermediate tendon, running from the scapula to the hyoid bone?

- **(A)** Omohyoid
- **(B)** Digastric
- **(C)** Sternohyoid
- **(D)** Thyrohyoid
- **(E)** Stylohyoid

17. The costoclavicular space, or cervicoaxillary canal, lies between the clavicle and which structure?

- **(A)** Coracoid process
- **(B)** First rib
- **(C)** Scalenus anterior
- **(D)** Manubrium
- **(E)** Second rib

02 SCALP, FACE, PAROTID & CRANIAL FOSSAE — LECTURES 3-4

18. Which muscle of facial expression forms an ellipse around the mouth and closes the lips?

- **(A)** Risorius
- **(B)** Depressor labii inferioris
- **(C)** Orbicularis oris
- **(D)** Buccinator
- **(E)** Mentalis

19. Which muscle draws the corner of the mouth upward and laterally, arising from the zygomatic bone?

- **(A)** Zygomaticus major
- **(B)** Levator labii superioris
- **(C)** Zygomaticus minor
- **(D)** Risorius
- **(E)** Levator anguli oris

20. Which muscle raises the corner of the mouth and helps form the nasolabial furrow, arising from the maxilla below the infraorbital foramen?

- **(A)** Zygomaticus major
- **(B)** Levator anguli oris
- **(C)** Risorius
- **(D)** Mentalis
- **(E)** Depressor anguli oris

21. Which muscle draws the corner of the mouth downward and laterally, arising from the oblique line of the mandible?

- **(A)** Mentalis
- **(B)** Levator anguli oris
- **(C)** Buccinator
- **(D)** Depressor anguli oris
- **(E)** Zygomaticus minor

22. Which muscle raises and protrudes the lower lip while wrinkling the skin of the chin?

- **(A)** Risorius
- **(B)** Depressor labii inferioris
- **(C)** Depressor anguli oris
- **(D)** Orbicularis oris
- **(E)** Mentalis

23. Risorius is unusual among the muscles of facial expression in arising from which structure, rather than directly from bone?

- **(A)** The zygomatic bone directly
- **(B)** Fascia over the masseter muscle
- **(C)** The mandible directly
- **(D)** The maxilla directly
- **(E)** The temporal bone directly

24. All the muscles of facial expression converge and intersect near the corner of the mouth at a point called what?

- **(A)** Glabella
- **(B)** Nasion
- **(C)** Pterion
- **(D)** Modiolus
- **(E)** Stomion

25. Which muscle arises from the frontal process of the maxilla and inserts on both the alar cartilage of the nose and the upper lip?

- **(A)** Depressor septi nasi muscle
- **(B)** Levator labii superioris alone
- **(C)** Zygomaticus minor muscle
- **(D)** Levator labii superioris alaeque nasi
- **(E)** Levator anguli oris muscle

26. Which muscle draws the lower lip downward and laterally, arising from the anterior part of the oblique line of the mandible?

- **(A)** Orbicularis oris
- **(B)** Risorius
- **(C)** Mentalis
- **(D)** Depressor anguli oris
- **(E)** Depressor labii inferioris

27. Which muscle raises the upper lip and helps form the nasolabial furrow, arising from the infraorbital margin of the maxilla?

- **(A)** Levator anguli oris
- **(B)** Orbicularis oris
- **(C)** Levator labii superioris
- **(D)** Levator labii superioris alaeque nasi
- **(E)** Zygomaticus major

28. The tarsofascial layer of the eyelid includes the tarsus and which other structure?

- **(A)** Orbital septum
- **(B)** Orbicularis oculi only
- **(C)** Bulbar conjunctiva only
- **(D)** Palpebral conjunctiva only
- **(E)** Levator palpebrae tendon only

29. The conjunctiva lining the eyelid is divided into which two parts?

- **(A)** Palpebral and bulbar conjunctiva
- **(B)** Anterior and posterior conjunctiva
- **(C)** Medial and lateral conjunctiva
- **(D)** Superior and inferior conjunctiva only
- **(E)** Tarsal and septal conjunctiva

30. Which part of orbicularis oculi closes the eyelids forcefully, forming an uninterrupted ellipse around the orbit?

- **(A)** Lacrimal part only
- **(B)** Orbital part
- **(C)** Tarsal part only
- **(D)** Palpebral part
- **(E)** Ciliary part

31. Which part of orbicularis oculi closes the eyelids gently, running from the medial to the lateral palpebral ligament?

- **(A)** Palpebral part
- **(B)** Tarsal part only
- **(C)** Orbital part
- **(D)** Ciliary part
- **(E)** Lacrimal part only

32. Which small muscle draws the eyebrows medially and downward, producing vertical frown lines?

- **(A)** Orbicularis oculi
- **(B)** Procerus
- **(C)** Depressor supercilii alone
- **(D)** Frontalis
- **(E)** Corrugator supercilii

33. The muscles of facial expression are classified into functional groups. Which group includes orbicularis oculi and corrugator supercilii?

- **(A)** Auricular group
- **(B)** Scalp group
- **(C)** Oral group
- **(D)** Nasal group
- **(E)** Orbital group

34. Which muscle draws down the medial angle of the eyebrows, producing transverse wrinkles over the bridge of the nose?

- **(A)** Procerus
- **(B)** Depressor septi
- **(C)** Orbicularis oculi
- **(D)** Corrugator supercilii
- **(E)** Nasalis

35. Which muscle, via its transverse part, compresses the nasal aperture?

- **(A)** Depressor septi
- **(B)** Procerus
- **(C)** Orbicularis oris
- **(D)** Levator labii superioris alaeque nasi
- **(E)** Nasalis

36. Which muscle, via its alar part, draws the alar cartilage downward and laterally to open the nostril?

- **(A)** Depressor septi
- **(B)** Corrugator supercilii
- **(C)** Levator labii superioris alaeque nasi
- **(D)** Nasalis
- **(E)** Procerus

37. Which small muscle pulls the nose inferiorly, arising from the maxilla above the medial incisor?

- **(A)** Depressor septi
- **(B)** Orbicularis oris
- **(C)** Nasalis (transverse part)
- **(D)** Procerus
- **(E)** Corrugator supercilii

03 DEVELOPMENT OF HEAD AND NECK — LECTURE 5

38. The first pharyngeal pouch gives rise to which structure?

- **(A)** Ultimobranchial body
- **(B)** Inferior parathyroid gland
- **(C)** Auditory (pharyngotympanic) tube
- **(D)** Palatine tonsil
- **(E)** Thymus gland

39. Which pharyngeal arch gives rise to the muscles of facial expression?

- **(A)** First pharyngeal arch
- **(B)** Fifth pharyngeal arch
- **(C)** Third pharyngeal arch
- **(D)** Second pharyngeal arch
- **(E)** Fourth pharyngeal arch

40. Neural crest cells contributing to the pharyngeal arches arise from which region of the developing embryo?

- **(A)** The somites of the trunk
- **(B)** The caudal primitive streak
- **(C)** The notochordal plate
- **(D)** The intraembryonic coelom
- **(E)** Lateral to the rostral neural tube

41. Which pharyngeal arch cartilage is known as Meckel's cartilage?

- **(A)** Fourth pharyngeal arch cartilage
- **(B)** First pharyngeal arch cartilage
- **(C)** Third pharyngeal arch cartilage
- **(D)** Second pharyngeal arch cartilage
- **(E)** Fifth pharyngeal arch cartilage

42. Which pharyngeal arch cartilage is known as Reichert's cartilage?

- **(A)** Second pharyngeal arch cartilage
- **(B)** First pharyngeal arch cartilage
- **(C)** Fourth pharyngeal arch cartilage
- **(D)** Third pharyngeal arch cartilage
- **(E)** Fifth pharyngeal arch cartilage

43. Only the first pharyngeal groove and first pharyngeal membrane normally persist in the adult. What happens to the second, third, and fourth grooves?

- **(A)** They are obliterated as the second arch overgrows them
- **(B)** They persist to form the external ear canal
- **(C)** They form the middle ear cavity
- **(D)** They become the palatine tonsil
- **(E)** They fuse to form the thyroid gland

44. Branchial is used as an older, alternative term for which word in this developmental system?

- **(A)** Cervical
- **(B)** Cardiac
- **(C)** Cranial
- **(D)** Pharyngeal
- **(E)** Neural

45. Failure of neural crest cells to migrate properly into the pharyngeal arches is associated with which type of congenital anomaly?

- **(A)** Renal agenesis
- **(B)** Cardiac septal defects only
- **(C)** Neural tube defects
- **(D)** Limb reduction defects only
- **(E)** Pharyngeal arch syndromes

46. A branchial cyst is best described as a remnant of which structure?

- **(A)** Thyroglossal duct
- **(B)** Notochord
- **(C)** Cloacal membrane
- **(D)** First pharyngeal pouch
- **(E)** Cervical sinus

47. The pharyngeal apparatus first becomes visible in the embryo during which week of development?

- **(A)** Eighth week
- **(B)** Fourth week
- **(C)** Second week
- **(D)** Sixth week
- **(E)** Twelfth week

48. Each pharyngeal arch contains local mesenchyme together with which additional cell population that expands the arch?

(A) Splanchnic mesoderm only
(B) Neural crest cells
(C) Endodermal cells only
(D) Ectodermal cells only
(E) Somatic mesoderm only

49. Each pharyngeal arch is said to keep the nerve of the arch from which it developed. Which nerve is associated with the second pharyngeal arch?

(A) Glossopharyngeal nerve
(B) Vagus nerve
(C) Trigeminal nerve
(D) Hypoglossal nerve
(E) Facial nerve

50. Which nerve is associated with the third pharyngeal arch?

(A) Trigeminal nerve
(B) Facial nerve
(C) Glossopharyngeal nerve
(D) Accessory nerve
(E) Vagus nerve

51. Which nerve is associated with the fourth and sixth pharyngeal arches?

(A) Hypoglossal nerve
(B) Glossopharyngeal nerve
(C) Vagus nerve
(D) Trigeminal nerve
(E) Facial nerve

52. Which nerve is associated with the first pharyngeal arch, supplying the muscles of mastication?

(A) Trigeminal nerve
(B) Facial nerve
(C) Vagus nerve
(D) Hypoglossal nerve
(E) Glossopharyngeal nerve

04 TEMPORAL & INFRATEMPORAL FOSSA, JAW JOINT — LECTURE 6

53. Which movement of the mandible is produced by bilateral contraction of the lateral pterygoid muscles?

(A) Retrusion
(B) Protrusion
(C) Depression by gravity alone
(D) Lateral excursion only
(E) Elevation

54. Which movement of the mandible is produced by the posterior fibres of temporalis contracting bilaterally?

(A) Depression
(B) Protrusion
(C) Rotation only
(D) Retrusion
(E) Lateral excursion

55. The medial pterygoid muscle has a deep head arising from which structure?

- **(A)** Coronoid process of the mandible
- **(B)** Medial surface of the lateral pterygoid plate
- **(C)** Maxillary tuberosity of the maxilla alone
- **(D)** Pterygoid fovea of the mandible
- **(E)** Zygomatic arch of the cheekbone

56. Which muscle inserts onto the medial surface of the ramus and angle of the mandible?

- **(A)** Medial pterygoid
- **(B)** Masseter
- **(C)** Temporalis
- **(D)** Lateral pterygoid
- **(E)** Buccinator

57. Which muscle inserts onto the lateral surface of the ramus and angle of the mandible?

- **(A)** Lateral pterygoid
- **(B)** Digastric
- **(C)** Medial pterygoid
- **(D)** Masseter
- **(E)** Temporalis

58. The superior head of lateral pterygoid arises from which bone?

- **(A)** Maxillary tuberosity
- **(B)** Greater wing of the sphenoid
- **(C)** Temporal bone
- **(D)** Lateral pterygoid plate
- **(E)** Zygomatic arch

59. Which nerve provides motor supply to all four muscles of mastication?

- **(A)** Facial nerve
- **(B)** Maxillary division of the trigeminal nerve
- **(C)** Glossopharyngeal nerve
- **(D)** Ophthalmic division of the trigeminal nerve
- **(E)** Mandibular division of the trigeminal nerve

60. Which nerve is at risk during an inferior alveolar nerve block, and supplies the lower teeth before emerging as the mental nerve?

- **(A)** Buccal nerve
- **(B)** Inferior alveolar nerve
- **(C)** Auriculotemporal nerve
- **(D)** Lingual nerve
- **(E)** Mylohyoid nerve

61. Which nerve provides the sole sensory supply to the buccal gingiva of the lower molar teeth?

- **(A)** Buccal nerve
- **(B)** Inferior alveolar nerve
- **(C)** Lingual nerve
- **(D)** Mental nerve
- **(E)** Auriculotemporal nerve

62. Which structure of the temporomandibular joint attaches peripherally to the joint capsule and is pulled forward by the lateral pterygoid?

(A) Synovial membrane alone
(B) Sphenomandibular ligament
(C) Stylomandibular ligament
(D) Articular disc
(E) Lateral ligament

63. Which artery is the larger terminal branch of the external carotid artery, supplying the infratemporal fossa?

(A) Superficial temporal artery
(B) Occipital artery
(C) Maxillary artery
(D) Facial artery
(E) Lingual artery

64. Which muscle of mastication is described as having a broad, fan-shaped origin from the temporal fossa?

(A) Buccinator
(B) Masseter
(C) Temporalis
(D) Medial pterygoid
(E) Lateral pterygoid

65. Which structure encircles the joint like a sleeve, attaching above to the articular tubercle and mandibular fossa, and below to the neck of the mandible?

(A) Articular capsule
(B) Synovial membrane alone
(C) Sphenomandibular ligament
(D) Stylomandibular ligament
(E) Articular disc

05 EXTRAOCULAR MUSCLES & INNERVATION — LECTURE 7

66. Which structure protects the cornea and eyeball from injury, along with the lacrimal fluid?

(A) Orbital septum alone
(B) Conjunctiva alone
(C) Extraocular muscles
(D) Optic nerve sheath
(E) Eyelids

67. Which gland secretes the lacrimal fluid that helps protect the cornea?

(A) Meibomian gland
(B) Submandibular gland
(C) Lacrimal gland
(D) Ciliary gland
(E) Sublingual gland

68. Which structure in the eyelid gives it its firm shape, described as dense connective tissue?

(A) Orbital septum
(B) Levator palpebrae tendon
(C) Conjunctiva
(D) Tarsus
(E) Orbicularis oculi

69. Which fibrous membrane of the eyelid helps limit the spread of infection between the eyelid and the orbit?

- **(A)** Tarsus
- **(B)** Levator aponeurosis
- **(C)** Orbital septum
- **(D)** Conjunctiva
- **(E)** Bulbar fascia

70. Which extraocular muscle originates at the apex of the orbit near the optic canal, but is not one of the four recti?

- **(A)** Inferior oblique
- **(B)** Superior oblique
- **(C)** Superior rectus
- **(D)** Lateral rectus
- **(E)** Levator palpebrae superioris only

71. Which nerve exits the orbit apex region and, unlike the other ocular motor nerves, passes outside the common tendinous ring?

- **(A)** Trochlear nerve
- **(B)** Facial nerve
- **(C)** Optic nerve
- **(D)** Abducens nerve
- **(E)** Oculomotor nerve

72. Which structure passes through the optic canal at the orbital apex, together with the optic nerve?

- **(A)** Abducens nerve
- **(B)** Oculomotor nerve
- **(C)** Trochlear nerve
- **(D)** Ophthalmic vein
- **(E)** Ophthalmic artery

73. Which nerve provides sensory innervation to the skin of the upper eyelid and forehead via its supraorbital branch?

- **(A)** Infraorbital nerve
- **(B)** Zygomatic nerve
- **(C)** Maxillary nerve
- **(D)** Facial nerve
- **(E)** Ophthalmic nerve

74. Which nerve gives rise to the infraorbital nerve, supplying the lower eyelid?

- **(A)** Maxillary nerve
- **(B)** Mandibular nerve
- **(C)** Ophthalmic nerve
- **(D)** Zygomaticofacial nerve alone
- **(E)** Facial nerve

75. Which motor nerve innervates orbicularis oculi, allowing forceful closure of the eyelids?

- **(A)** Trigeminal nerve
- **(B)** Abducens nerve
- **(C)** Facial nerve
- **(D)** Oculomotor nerve
- **(E)** Optic nerve

76. Which structure of the eye is responsible for producing tears that lubricate and protect the cornea?

- **(A)** Meibomian glands
- **(B)** Ciliary body
- **(C)** Sclera
- **(D)** Conjunctiva alone
- **(E)** Lacrimal gland

06 THORACIC WALL AND MUSCLES — LECTURE 8

77. The sternal part of the diaphragm arises from which structure?

- **(A)** Posterior aspect of the xiphoid process
- **(B)** Twelfth rib
- **(C)** Lower six costal cartilages
- **(D)** Medial and lateral arcuate ligaments
- **(E)** Body of the second lumbar vertebra

78. The costal part of the diaphragm arises from which structure?

- **(A)** Manubrium
- **(B)** Internal surfaces of the lower costal cartilages
- **(C)** Medial and lateral arcuate ligaments alone
- **(D)** Xiphoid process
- **(E)** Body of the first lumbar vertebra

79. The lumbar part of the diaphragm arises via the crura and which pair of ligaments?

- **(A)** Inguinal ligaments
- **(B)** Sacrotuberous ligaments
- **(C)** Medial and lateral arcuate ligaments
- **(D)** Costoclavicular ligaments
- **(E)** Suspensory ligaments

80. Which nerve provides the sole motor supply to the diaphragm?

- **(A)** Long thoracic nerve
- **(B)** Vagus nerve
- **(C)** Phrenic nerve
- **(D)** Intercostal nerves
- **(E)** Sympathetic trunk

81. A pneumothorax may result from injury to which structure surrounding the lungs?

- **(A)** Diaphragm alone
- **(B)** Mediastinum
- **(C)** Pleura
- **(D)** Pericardium
- **(E)** Endothoracic fascia alone

82. Which rib articulates with the body of only one vertebra, making it an atypical rib?

- **(A)** Fifth rib
- **(B)** Third rib
- **(C)** Ninth rib
- **(D)** First rib
- **(E)** Seventh rib

83. Scalenus anterior divides which artery into three named parts as it crosses the root of the neck?

- **(A)** Subclavian artery
- **(B)** Common carotid artery
- **(C)** Vertebral artery
- **(D)** External carotid artery
- **(E)** Internal thoracic artery

84. Which part of the subclavian artery lies lateral to scalenus anterior and continues as the axillary artery?

- **(A)** Fourth part
- **(B)** Second part
- **(C)** First part
- **(D)** Third part
- **(E)** There is no such division

85. The manubriosternal joint is classified as which type of joint?

- **(A)** Primary cartilaginous joint (synchondrosis)
- **(B)** Secondary cartilaginous joint (symphysis)
- **(C)** Syndesmosis
- **(D)** Synovial plane joint
- **(E)** Fibrous suture

86. Which muscle layer of the intercostal space assists in expiration by depressing the ribs?

- **(A)** Internal intercostal muscle (interosseous part)
- **(B)** Levator costarum muscle of the back
- **(C)** Pectoralis minor muscle of the chest
- **(D)** External intercostal muscle layer entirely
- **(E)** Serratus anterior muscle of the chest

87. Which structure is described as the innermost of the three intercostal muscle layers, lying deep to the neurovascular bundle?

- **(A)** Internal intercostal muscle
- **(B)** Transversus thoracis
- **(C)** Innermost intercostal muscle
- **(D)** External intercostal muscle
- **(E)** Levator costarum

88. Which structure lies in the plane between the internal intercostal and innermost intercostal muscles?

- **(A)** The pericardium
- **(B)** The intercostal neurovascular bundle
- **(C)** The endothoracic fascia alone, with no vessels
- **(D)** The parietal pleura directly
- **(E)** The visceral pleura

89. Which structure of the thoracic wall is a thin layer of connective tissue lining the internal surface, between the innermost intercostal muscles and the parietal pleura?

- **(A)** Investing fascia
- **(B)** Buccopharyngeal fascia
- **(C)** Pretracheal fascia
- **(D)** Endothoracic fascia
- **(E)** Prevertebral fascia

07 ANTERIOR ABDOMINAL WALL & HERNIA

LECTURE 9

90. Which muscle layer is the innermost of the three flat muscles of the anterolateral abdominal wall?

- **(A)** Pyramidalis
- **(B)** Internal oblique
- **(C)** Transversus abdominis
- **(D)** Rectus abdominis
- **(E)** External oblique

91. Which muscle layer is the outermost of the three flat muscles of the anterolateral abdominal wall?

- **(A)** Transversus abdominis
- **(B)** Rectus abdominis
- **(C)** External oblique
- **(D)** Transversalis fascia
- **(E)** Internal oblique

92. Which layer of the abdominal wall lines the inner aspect of the cavity, deep to transversus abdominis?

- **(A)** Peritoneum directly
- **(B)** Rectus sheath
- **(C)** Camper fascia
- **(D)** Scarpa fascia
- **(E)** Transversalis fascia

93. The superficial fascia of the lower anterior abdominal wall splits into two layers. What is the deeper, membranous layer called?

- **(A)** Colles fascia
- **(B)** Scarpa fascia
- **(C)** Camper fascia
- **(D)** Transversalis fascia
- **(E)** Buck fascia

94. Scarpa fascia fuses to the deep fascia of the thigh at which structure, preventing extravasated fluid from tracking into the thigh?

- **(A)** The linea alba
- **(B)** The pubic symphysis
- **(C)** Just below the inguinal ligament
- **(D)** The anterior superior iliac spine
- **(E)** The umbilicus

95. Which two muscles form the posterior abdominal wall medially, alongside quadratus lumborum more laterally?

- **(A)** Diaphragm and psoas minor
- **(B)** Transversus abdominis and rectus abdominis
- **(C)** Psoas major and iliacus
- **(D)** Rectus abdominis and pyramidalis
- **(E)** External oblique and internal oblique

96. The abdominal cavity is bounded superiorly by which structure?

- **(A)** Diaphragm
- **(B)** Pelvic inlet
- **(C)** Twelfth rib
- **(D)** Iliac crest
- **(E)** Inguinal ligament

97. Which nerve, the anterior ramus of the twelfth thoracic spinal nerve, runs along the inferior border of the twelfth rib?

- **(A)** Iliohypogastric nerve
- **(B)** Lateral cutaneous branch of the eleventh nerve
- **(C)** Ilioinguinal nerve
- **(D)** Genitofemoral nerve
- **(E)** Subcostal nerve

98. Which nerve is the superior terminal branch of the first lumbar anterior ramus, supplying skin over the iliac crest?

- **(A)** Genitofemoral nerve
- **(B)** Ilioinguinal nerve
- **(C)** Subcostal nerve
- **(D)** Iliohypogastric nerve
- **(E)** Femoral nerve

99. Which nerve is the inferior terminal branch of the first lumbar anterior ramus, and unlike its companion nerve, actually traverses the inguinal canal?

- **(A)** Subcostal nerve
- **(B)** Ilioinguinal nerve
- **(C)** Iliohypogastric nerve
- **(D)** Genitofemoral nerve
- **(E)** Obturator nerve

100. The rectus sheath encloses which muscle along most of its length?

- **(A)** Transversus abdominis
- **(B)** Rectus abdominis
- **(C)** External oblique
- **(D)** Internal oblique
- **(E)** Quadratus lumborum

101. The linea alba runs in the midline from the xiphoid process to which structure below?

- **(A)** Inguinal ligament
- **(B)** Anterior superior iliac spine
- **(C)** Umbilicus only
- **(D)** Pubic tubercle
- **(E)** Pubic symphysis

102. Which structure marks the boundary above which the posterior rectus sheath is present, and below which it is absent?

- **(A)** Linea semilunaris
- **(B)** Inguinal ligament
- **(C)** Linea alba
- **(D)** Umbilicus
- **(E)** Arcuate line

08 MUSCLE PHYSIOLOGY — LECTURES 10-12

103. The sliding filament mechanism describes contraction as resulting from which process?

- **(A)** Thick filaments shortening in length
- **(B)** Thin filaments shortening in length
- **(C)** Both filament types shortening equally
- **(D)** Thick and thin filaments sliding past each other
- **(E)** The Z discs dissolving during contraction

104. During muscle shortening according to the sliding filament mechanism, what happens to the length of the individual filaments themselves?

- **(A)** Both filaments shorten equally
- **(B)** Both filaments lengthen
- **(C)** Neither filament changes length
- **(D)** Only the thick filaments shorten
- **(E)** Only the thin filaments shorten

105. Which zone of the sarcomere narrows as thin filaments slide further into the region between thick filaments during contraction?

- **(A)** Z disc
- **(B)** Distance between A bands
- **(C)** M line width
- **(D)** H zone
- **(E)** A band

106. Which motor unit type is recruited first when a muscle contraction begins, according to the size principle?

- **(A)** Only fast-twitch units respond to any stimulus
- **(B)** Large, fast-twitch motor units
- **(C)** Both types simultaneously
- **(D)** Small, slow-twitch motor units
- **(E)** Units are recruited randomly regardless of size

107. Slow-twitch motor units are characteristically small and control which type of movement?

- **(A)** Powerful, gross movements only
- **(B)** Fine, precise movements
- **(C)** No voluntary movements
- **(D)** Only involuntary movements
- **(E)** Only reflex movements

108. Which structure of the neuromuscular junction contains vesicles of acetylcholine, forming the presynaptic portion?

- **(A)** Postjunctional folds
- **(B)** Motor end plate as a whole
- **(C)** Synaptic cleft
- **(D)** Axon terminal
- **(E)** Sarcoplasmic reticulum

109. Which structure of the neuromuscular junction bears the nicotinic acetylcholine receptors, forming the postsynaptic portion?

- **(A)** Schwann cell sheath
- **(B)** Axon terminal
- **(C)** Synaptic vesicles
- **(D)** Postjunctional folds
- **(E)** Presynaptic membrane

110. What is the narrow space called that separates the nerve terminal from the muscle fibre at the neuromuscular junction?

- **(A)** Synaptic cleft
- **(B)** Subneural cleft only
- **(C)** Sarcoplasmic reticulum
- **(D)** Terminal cisterna
- **(E)** Transverse tubule

111. A greater frequency of action potentials reaching the motor axon terminal produces which effect at the neuromuscular junction?

- **(A)** A complete blockade of the postsynaptic receptor site
- **(B)** An immediate depletion of all the synaptic vesicles
- **(C)** A higher concentration of acetylcholine in the synaptic cleft
- **(D)** No change at all in the acetylcholine concentration
- **(E)** A lower concentration of acetylcholine being released

112. Myasthenia gravis exists in two major clinical forms. Which form involves weakness of only the extraocular muscles?

- **(A)** Bulbar myasthenia gravis only
- **(B)** Generalized myasthenia gravis
- **(C)** Late-onset myasthenia gravis only
- **(D)** Ocular myasthenia gravis
- **(E)** Congenital myasthenia gravis only

113. In myasthenia gravis, anti-acetylcholine receptor antibodies cause degeneration of which structure?

- **(A)** Presynaptic axon terminal
- **(B)** Muscle spindle
- **(C)** Sarcoplasmic reticulum
- **(D)** Transverse tubules
- **(E)** Postjunctional folds

114. Which structure senses depolarisation of the transverse tubule and mechanically opens the calcium release channel of the sarcoplasmic reticulum?

- **(A)** Calmodulin
- **(B)** Ryanodine receptor alone
- **(C)** Nicotinic acetylcholine receptor
- **(D)** Troponin C
- **(E)** Dihydropyridine receptor

115. Which receptor of the sarcoplasmic reticulum is directly opened by the dihydropyridine receptor to release calcium?

- **(A)** Calmodulin receptor
- **(B)** Nicotinic acetylcholine receptor
- **(C)** Voltage-gated sodium channel
- **(D)** Troponin C
- **(E)** Ryanodine receptor

116. Once released into the cytoplasm, calcium is taken back up by which structure to allow relaxation?

- **(A)** Nucleus
- **(B)** Extracellular space directly
- **(C)** Mitochondria primarily
- **(D)** Sarcoplasmic reticulum
- **(E)** Golgi apparatus

117. Which molecule directly binds to the myosin head to allow it to detach from actin during the cross-bridge cycle?

- **(A)** Adenosine triphosphate
- **(B)** Adenosine diphosphate
- **(C)** Troponin
- **(D)** Calcium ion
- **(E)** Tropomyosin

118. Hydrolysis of adenosine triphosphate on the myosin head provides the energy for which step of the cross-bridge cycle?

- **(A)** Binding of calcium to troponin
- **(B)** Re-cocking the myosin head for the next power stroke
- **(C)** Release of acetylcholine
- **(D)** Reuptake of calcium by the sarcoplasmic reticulum
- **(E)** Depolarisation of the transverse tubule

119. Which ion, once bound to troponin C, moves tropomyosin away from the myosin-binding sites on actin?

- **(A)** Sodium
- **(B)** Calcium
- **(C)** Chloride
- **(D)** Potassium
- **(E)** Magnesium

120. Which part of the sarcomere represents the full length of the thick filaments and stays constant in width during contraction?

- **(A)** A band
- **(B)** M line width
- **(C)** H zone
- **(D)** I band
- **(E)** Z disc to Z disc distance

Answer Key & Reasoning

Correct answer in green, with the reasoning beneath

Each answer was checked before this paper was issued, so you should not need a second opinion.

01 NECK: TRIANGLES, VESSELS & NERVES

LECTURES 1-2

1. The posterior triangle of the neck is bounded anteriorly by which muscle?

D — Sternocleidomastoid

The posterior triangle is bounded anteriorly by the posterior border of sternocleidomastoid, posteriorly by the anterior border of trapezius, and inferiorly by the middle third of the clavicle.

2. Which triangle of the neck lies within the lower part of the posterior triangle and contains the third part of the subclavian artery?

C — Subclavian triangle

The subclavian triangle is the smaller, inferior subdivision of the posterior triangle, separated from the occipital triangle above by the inferior belly of omohyoid, and it contains the third part of the subclavian artery.

3. Which muscle divides the posterior triangle into an upper occipital triangle and a lower subclavian triangle?

C — Inferior belly of omohyoid

The inferior belly of omohyoid crosses the posterior triangle and divides it into the occipital triangle above and the subclavian triangle below.

4. Which nerve descends on the anterior surface of scalenus anterior to supply the diaphragm?

E — Phrenic nerve

The phrenic nerve descends on the anterior surface of scalenus anterior, deep to the prevertebral fascia, before entering the thorax to supply the diaphragm.

5. The submandibular gland is bounded by the two bellies of which muscle?

C — Digastric

The submandibular triangle, containing the submandibular gland, is bounded by the anterior and posterior bellies of digastric and the lower border of the mandible.

6. Which triangle of the neck is bounded by the two anterior bellies of digastric and the hyoid bone?

E — Submental triangle

The submental triangle lies beneath the chin, bounded by the two anterior bellies of digastric and the hyoid bone below.

7. Which vein passes through the substance of the parotid gland before joining the external jugular system?

D — Retromandibular vein

The retromandibular vein forms within the parotid gland from the union of the superficial temporal and maxillary veins, then divides into anterior and posterior branches as it leaves the gland.

8. Which nerve supplies the sternohyoid, sternothyroid, and omohyoid muscles?

B — Ansa cervicalis

The ansa cervicalis, a loop formed from cervical plexus fibres, supplies sternohyoid, sternothyroid, and both bellies of omohyoid, the infrahyoid strap muscles.

9. Which muscle is the exception among the infrahyoid muscles, receiving direct first cervical fibres via the hypoglossal nerve rather than the ansa cervicalis?

B — Thyrohyoid

Thyrohyoid is supplied by first cervical fibres that travel with the hypoglossal nerve and branch off separately, unlike the other infrahyoid muscles which are supplied by the ansa cervicalis.

10. Which nerve crosses superficial to the internal and external carotid arteries to reach the tongue?

B — Hypoglossal nerve

The hypoglossal nerve crosses superficial to both the internal and external carotid arteries high in the neck before turning forward to supply the muscles of the tongue.

11. Which artery is the first branch of the external carotid artery?

A — Superior thyroid artery

The superior thyroid artery is the first and most inferior branch of the external carotid artery, supplying the upper pole of the thyroid gland and the larynx.

12. The internal jugular vein terminates by joining the subclavian vein to form which structure?

D — Brachiocephalic vein

The internal jugular vein joins the subclavian vein behind the sternoclavicular joint to form the brachiocephalic vein on each side.

13. Within the carotid sheath, which nerve runs the whole length of the neck alongside the great vessels?

E — Vagus nerve

The vagus nerve runs the length of the carotid sheath in the groove between the internal jugular vein and the carotid artery.

14. The thyroid gland is enclosed by which layer of deep cervical fascia?

C — Pretracheal fascia

The pretracheal layer of deep cervical fascia encloses the thyroid gland, trachea, and oesophagus, and is why the thyroid moves with swallowing.

15. Omohyoid has two bellies joined by an intermediate tendon. Which belly lies closer to the hyoid bone?

A — Superior belly

The superior belly of omohyoid lies closer to the hyoid bone, running from the intermediate tendon up to the hyoid, while the inferior belly runs from the scapula to the intermediate tendon lower in the neck.

16. Which muscle has two bellies connected by an intermediate tendon, running from the scapula to the hyoid bone?

A — Omohyoid

Omohyoid is a two-bellied muscle connected by an intermediate tendon, running from the superior border of the scapula to the hyoid bone, and it helps subdivide the anterior and posterior triangles of the neck.

17. The costoclavicular space, or cervicoaxillary canal, lies between the clavicle and which structure?

B — First rib

The costoclavicular space, also called the cervicoaxillary canal, lies between the clavicle and the first rib, and structures passing through it are at risk of compression in thoracic outlet syndrome.

02 SCALP, FACE, PAROTID & CRANIAL FOSSAE

LECTURES 3-4

18. Which muscle of facial expression forms an ellipse around the mouth and closes the lips?

C — Orbicularis oris

Orbicularis oris forms an ellipse of fibres around the mouth, arising from muscles in the surrounding area, and its action closes and protrudes the lips.

19. Which muscle draws the corner of the mouth upward and laterally, arising from the zygomatic bone?

A — Zygomaticus major

Zygomaticus major arises from the posterior part of the lateral surface of the zygomatic bone and inserts at the corner of the mouth, drawing it upward and laterally, the main muscle of smiling.

20. Which muscle raises the corner of the mouth and helps form the nasolabial furrow, arising from the maxilla below the infraorbital foramen?

B — Levator anguli oris

Levator anguli oris arises from the maxilla just below the infraorbital foramen and inserts at the corner of the mouth, raising it and contributing to the nasolabial furrow.

21. Which muscle draws the corner of the mouth downward and laterally, arising from the oblique line of the mandible?

D — Depressor anguli oris

Depressor anguli oris arises from the oblique line of the mandible below the canine, premolar, and first molar teeth, and draws the corner of the mouth down and laterally.

22. Which muscle raises and protrudes the lower lip while wrinkling the skin of the chin?

E — Mentalis

Mentalis arises from the mandible inferior to the incisor teeth and inserts into the skin of the chin, raising and protruding the lower lip while producing the characteristic wrinkled chin.

23. Risorius is unusual among the muscles of facial expression in arising from which structure, rather than directly from bone?

B — Fascia over the masseter muscle

Unlike most muscles of facial expression, which arise directly from bone, risorius arises from the fascia overlying masseter and inserts into the skin at the corner of the mouth, retracting it.

24. All the muscles of facial expression converge and intersect near the corner of the mouth at a point called what?

D — Modiolus

The modiolus is the fibromuscular junction near the corner of the mouth where several muscles of facial expression, including orbicularis oris, buccinator, and the levators and depressors, converge and interweave.

25. Which muscle arises from the frontal process of the maxilla and inserts on both the alar cartilage of the nose and the upper lip?

D — Levator labii superioris alaeque nasi

Levator labii superioris alaeque nasi arises from the frontal process of the maxilla and inserts on both the alar cartilage of the nose and the upper lip, raising the upper lip and opening the nostril at the same time.

26. Which muscle draws the lower lip downward and laterally, arising from the anterior part of the oblique line of the mandible?

E — Depressor labii inferioris

Depressor labii inferioris arises from the anterior part of the oblique line of the mandible and inserts into the lower lip at the midline, drawing the lower lip downward and laterally.

27. Which muscle raises the upper lip and helps form the nasolabial furrow, arising from the infraorbital margin of the maxilla?

C — Levator labii superioris

Levator labii superioris arises from the infraorbital margin of the maxilla and inserts into the skin of the upper lateral lip, raising the upper lip and helping form the nasolabial furrow.

28. The tarsofascial layer of the eyelid includes the tarsus and which other structure?

A — Orbital septum

The tarsofascial layer of the eyelid consists of the tarsus, a plate of dense connective tissue giving the lid its shape, and the orbital septum, a fibrous membrane that helps limit the spread of infection.

29. The conjunctiva lining the eyelid is divided into which two parts?

A — Palpebral and bulbar conjunctiva

The conjunctiva is divided into the palpebral conjunctiva, lining the inner surface of the eyelids, and the bulbar conjunctiva, covering the visible white of the eye, meeting at the conjunctival fornix.

30. Which part of orbicularis oculi closes the eyelids forcefully, forming an uninterrupted ellipse around the orbit?

B — Orbital part

The orbital part of orbicularis oculi forms an uninterrupted ellipse of fibres around the orbit, arising from the frontal bone, maxilla, and medial palpebral ligament, and it closes the eyelids forcefully.

31. Which part of orbicularis oculi closes the eyelids gently, running from the medial to the lateral palpebral ligament?

A — Palpebral part

The palpebral part of orbicularis oculi runs from the medial palpebral ligament to the lateral palpebral raphe, producing gentle, involuntary closure of the eyelids, as in blinking.

32. Which small muscle draws the eyebrows medially and downward, producing vertical frown lines?

E — Corrugator supercilii

Corrugator supercilii arises from the medial end of the superciliary arch and inserts into the skin of the medial eyebrow, drawing the eyebrows medially and downward to produce vertical frown lines.

33. The muscles of facial expression are classified into functional groups. Which group includes orbicularis oculi and corrugator supercilii?

E — Orbital group

The orbital group of facial expression muscles includes orbicularis oculi and corrugator supercilii, both acting on the eyelids and eyebrows.

34. Which muscle draws down the medial angle of the eyebrows, producing transverse wrinkles over the bridge of the nose?

A — Procerus

Procerus arises from the nasal bone and upper lateral nasal cartilage and inserts into the skin of the lower forehead between the eyebrows, producing transverse wrinkles over the nasal bridge.

35. Which muscle, via its transverse part, compresses the nasal aperture?

E — Nasalis

The transverse part of nasalis arises from the maxilla just lateral to the nose and inserts into an aponeurosis across the dorsum of the nose, compressing the nasal aperture.

36. Which muscle, via its alar part, draws the alar cartilage downward and laterally to open the nostril?

D — Nasalis

The alar part of nasalis arises from the maxilla over the lateral incisor and inserts into the alar cartilage of the nose, drawing it downward and laterally to open the nostril.

37. Which small muscle pulls the nose inferiorly, arising from the maxilla above the medial incisor?

A — Depressor septi

Depressor septi arises from the maxilla above the medial incisor and inserts into the mobile part of the nasal septum, pulling the nose inferiorly.

03 DEVELOPMENT OF HEAD AND NECK — LECTURE 5

38. The first pharyngeal pouch gives rise to which structure?

C — Auditory (pharyngotympanic) tube

The first pharyngeal pouch expands to form the auditory tube and contributes to the middle ear cavity, remaining connected to the pharynx throughout life as the pharyngotympanic tube.

39. Which pharyngeal arch gives rise to the muscles of facial expression?

D — Second pharyngeal arch

The second pharyngeal arch gives rise to the muscles of facial expression, supplied by the facial nerve, the second arch's own nerve.

40. Neural crest cells contributing to the pharyngeal arches arise from which region of the developing embryo?

E — Lateral to the rostral neural tube

Neural crest cells destined for the pharyngeal arches arise lateral to the rostral part of the developing neural tube and migrate into each arch, where they proliferate alongside the local mesenchyme.

41. Which pharyngeal arch cartilage is known as Meckel's cartilage?

B — First pharyngeal arch cartilage

Meckel's cartilage is the cartilage of the first pharyngeal arch, giving rise to the malleus and incus of the middle ear.

42. Which pharyngeal arch cartilage is known as Reichert's cartilage?

A — Second pharyngeal arch cartilage

Reichert's cartilage is the cartilage of the second pharyngeal arch, giving rise to the stapes, the styloid process, and part of the hyoid bone.

43. Only the first pharyngeal groove and first pharyngeal membrane normally persist in the adult. What happens to the second, third, and fourth grooves?

A — They are obliterated as the second arch overgrows them

As the second pharyngeal arch grows caudally, it overgrows and buries the second, third, and fourth grooves within a temporary space called the cervical sinus, which normally disappears completely.

44. Branchial is used as an older, alternative term for which word in this developmental system?

D — Pharyngeal

Branchial is used interchangeably with pharyngeal to describe the arches, pouches, grooves, and membranes of this developmental system, reflecting the older terminology derived from the gill (branchial) apparatus of lower vertebrates.

45. Failure of neural crest cells to migrate properly into the pharyngeal arches is associated with which type of congenital anomaly?

E — Pharyngeal arch syndromes

Failure of neural crest cells to migrate into the pharyngeal arches is a recognised cause of pharyngeal arch syndromes, producing a range of craniofacial anomalies depending on which arch and structures are affected.

46. A branchial cyst is best described as a remnant of which structure?

E — Cervical sinus

A branchial cyst is a remnant of the cervical sinus, the temporary space formed when the second pharyngeal arch overgrows the lower arches, and it typically presents as a swelling along the anterior border of sternocleidomastoid.

47. The pharyngeal apparatus first becomes visible in the embryo during which week of development?

B — Fourth week

The pharyngeal apparatus, comprising the arches, pouches, grooves, and membranes, first becomes observable in the embryo during the fourth week of development.

48. Each pharyngeal arch contains local mesenchyme together with which additional cell population that expands the arch?

B — Neural crest cells

Each pharyngeal arch contains both local mesenchyme and invading neural crest cells, and the arch expands through proliferation of these neural crest cells alongside the resident mesenchyme.

49. Each pharyngeal arch is said to keep the nerve of the arch from which it developed. Which nerve is associated with the second pharyngeal arch?

E — Facial nerve

The second pharyngeal arch is supplied by the facial nerve, and all structures derived from this arch, including the muscles of facial expression, retain this nerve regardless of where they migrate.

50. Which nerve is associated with the third pharyngeal arch?

C — Glossopharyngeal nerve

The third pharyngeal arch is supplied by the glossopharyngeal nerve, and stylopharyngeus, the sole muscle derived from this arch, is correspondingly innervated by it.

51. Which nerve is associated with the fourth and sixth pharyngeal arches?

C — Vagus nerve

The fourth and sixth pharyngeal arches are both supplied by the vagus nerve, consistent with their contribution to laryngeal and pharyngeal muscles innervated by vagal branches.

52. Which nerve is associated with the first pharyngeal arch, supplying the muscles of mastication?

A — Trigeminal nerve

The first pharyngeal arch is supplied by the trigeminal nerve, and all first-arch derivatives, including the muscles of mastication, retain this nerve supply.

04 TEMPORAL & INFRATEMPORAL FOSSA, JAW JOINT

LECTURE 6

53. Which movement of the mandible is produced by bilateral contraction of the lateral pterygoid muscles?

B — Protrusion

Bilateral contraction of the lateral pterygoid muscles produces protrusion of the mandible, pulling the condyles and disc forward onto the articular tubercle.

54. Which movement of the mandible is produced by the posterior fibres of temporalis contracting bilaterally?

D — Retrusion

The posterior, more horizontally oriented fibres of temporalis produce retrusion of the mandible when contracting bilaterally, pulling the jaw backward.

55. The medial pterygoid muscle has a deep head arising from which structure?

B — Medial surface of the lateral pterygoid plate

The deep head of medial pterygoid arises from the medial surface of the lateral pterygoid plate, while a smaller superficial head arises from the maxillary tuberosity.

56. Which muscle inserts onto the medial surface of the ramus and angle of the mandible?

A — Medial pterygoid

Medial pterygoid inserts onto the medial surface of the ramus and angle of the mandible, working with masseter on the opposite (lateral) surface to form a muscular sling that elevates the jaw.

57. Which muscle inserts onto the lateral surface of the ramus and angle of the mandible?

D — Masseter

Masseter inserts onto the lateral surface of the ramus and angle of the mandible, forming, together with medial pterygoid on the medial surface, a sling that elevates the jaw.

58. The superior head of lateral pterygoid arises from which bone?

B — Greater wing of the sphenoid

The superior head of lateral pterygoid arises from the greater wing of the sphenoid, while the larger inferior head arises from the lateral surface of the lateral pterygoid plate.

59. Which nerve provides motor supply to all four muscles of mastication?

E — Mandibular division of the trigeminal nerve

All four muscles of mastication are supplied by branches of the mandibular division of the trigeminal nerve, consistent with their shared first pharyngeal arch origin.

60. Which nerve is at risk during an inferior alveolar nerve block, and supplies the lower teeth before emerging as the mental nerve?

B — Inferior alveolar nerve

The inferior alveolar nerve enters the mandibular foramen to supply the lower teeth, later emerging through the mental foramen as the mental nerve to supply the chin and lower lip.

61. Which nerve provides the sole sensory supply to the buccal gingiva of the lower molar teeth?

A — Buccal nerve

The buccal nerve, the only sensory branch of the anterior division of the mandibular nerve, supplies the buccal gingiva of the lower molars, an area not covered by a standard inferior alveolar nerve block.

62. Which structure of the temporomandibular joint attaches peripherally to the joint capsule and is pulled forward by the lateral pterygoid?

D — Articular disc

The articular disc attaches peripherally to the joint capsule and receives fibres from the lateral pterygoid, so it is actively drawn forward with the condyle during protrusion and the opening movement.

63. Which artery is the larger terminal branch of the external carotid artery, supplying the infratemporal fossa?

C — Maxillary artery

The maxillary artery is the larger of the two terminal branches of the external carotid artery (the other being the superficial temporal artery), and it is the main arterial supply of the infratemporal fossa.

64. Which muscle of mastication is described as having a broad, fan-shaped origin from the temporal fossa?

C — Temporalis

Temporalis has a broad, fan-shaped origin from the floor of the temporal fossa, with its fibres converging as they descend to insert via a strong tendon on the coronoid process.

65. Which structure encircles the joint like a sleeve, attaching above to the articular tubercle and mandibular fossa, and below to the neck of the mandible?

A — Articular capsule

The articular capsule surrounds the temporomandibular joint like a sleeve, attaching above to the articular tubercle and margins of the mandibular fossa, and below to the neck of the mandible.

05 EXTRAOCULAR MUSCLES & INNERVATION

LECTURE 7

66. Which structure protects the cornea and eyeball from injury, along with the lacrimal fluid?

E — Eyelids

The eyelids, together with lacrimal fluid secreted by the lacrimal glands, protect the cornea and eyeball from injury and irritation.

67. Which gland secretes the lacrimal fluid that helps protect the cornea?

C — Lacrimal gland

The lacrimal gland secretes lacrimal fluid, which together with the eyelids protects the cornea and eyeball from injury and irritation.

68. Which structure in the eyelid gives it its firm shape, described as dense connective tissue?

D — Tarsus

The tarsus is a plate of dense connective tissue within the tarsofascial layer of the eyelid, giving it structural firmness and shape.

69. Which fibrous membrane of the eyelid helps limit the spread of infection between the eyelid and the orbit?

C — Orbital septum

The orbital septum is a fibrous membrane within the tarsofascial layer of the eyelid that acts as a barrier limiting the spread of infection between the superficial eyelid tissues and the deeper orbital contents.

70. Which extraocular muscle originates at the apex of the orbit near the optic canal, but is not one of the four recti?

B — Superior oblique

The superior oblique originates near the apex of the orbit, close to the optic canal, though it does not arise from the common tendinous ring itself like the four recti.

71. Which nerve exits the orbit apex region and, unlike the other ocular motor nerves, passes outside the common tendinous ring?

A — Trochlear nerve

The trochlear nerve enters the orbit through the superior orbital fissure but passes outside the common tendinous ring, unlike the oculomotor and abducens nerves which pass through it.

72. Which structure passes through the optic canal at the orbital apex, together with the optic nerve?

E — Ophthalmic artery

The ophthalmic artery passes through the optic canal together with the optic nerve, both sharing the same dural sheath as they enter the orbit.

73. Which nerve provides sensory innervation to the skin of the upper eyelid and forehead via its supraorbital branch?

E — Ophthalmic nerve

The ophthalmic division of the trigeminal nerve gives off the supraorbital nerve, which supplies sensation to the skin of the upper eyelid and forehead.

74. Which nerve gives rise to the infraorbital nerve, supplying the lower eyelid?

A — Maxillary nerve

The infraorbital nerve is a branch of the maxillary division of the trigeminal nerve, running in the floor of the orbit to supply the lower eyelid and adjacent skin of the face.

75. Which motor nerve innervates orbicularis oculi, allowing forceful closure of the eyelids?

C — Facial nerve

The facial nerve provides the sole motor innervation to orbicularis oculi, so its injury prevents forceful, and eventually even gentle, closure of the eyelids on the affected side.

76. Which structure of the eye is responsible for producing tears that lubricate and protect the cornea?

E — Lacrimal gland

The lacrimal gland produces the watery component of tears, which together with the eyelids protects the cornea and eyeball from injury and irritation.

06 THORACIC WALL AND MUSCLES

LECTURE 8

77. The sternal part of the diaphragm arises from which structure?

A — Posterior aspect of the xiphoid process

The sternal part of the diaphragm arises from the posterior aspect of the xiphoid process, one of the three muscular parts of origin along with the costal and lumbar parts.

78. The costal part of the diaphragm arises from which structure?

B — Internal surfaces of the lower costal cartilages

The costal part of the diaphragm arises from the internal surfaces of the lower costal cartilages and adjoining ribs, contributing the largest part of the muscular origin.

79. The lumbar part of the diaphragm arises via the crura and which pair of ligaments?

C — Medial and lateral arcuate ligaments

The lumbar part of the diaphragm arises from the crura and the medial and lateral arcuate ligaments, which arch over psoas major and quadratus lumborum respectively.

80. Which nerve provides the sole motor supply to the diaphragm?

C — Phrenic nerve

The phrenic nerve is the sole motor supply to the diaphragm, and its injury causes paralysis of the corresponding hemidiaphragm.

81. A pneumothorax may result from injury to which structure surrounding the lungs?

C — Pleura

A pneumothorax results from air entering the pleural cavity, typically through injury to the parietal or visceral pleura, causing the lung to collapse.

82. Which rib articulates with the body of only one vertebra, making it an atypical rib?

D — First rib

The first rib is atypical, articulating with the body of only the first thoracic vertebra through a single facet, rather than the two demifacets of a typical rib.

83. Scalenus anterior divides which artery into three named parts as it crosses the root of the neck?

A — Subclavian artery

The subclavian artery is divided into three parts by the scalenus anterior muscle: the first part lies medial to the muscle, the second part lies posterior to it, and the third part lies lateral to it.

84. Which part of the subclavian artery lies lateral to scalenus anterior and continues as the axillary artery?

D — Third part

The third part of the subclavian artery lies lateral to scalenus anterior, resting on the first rib, and continues beyond the outer border of the first rib as the axillary artery.

85. The manubriosternal joint is classified as which type of joint?

B — Secondary cartilaginous joint (symphysis)

The manubriosternal joint is a secondary cartilaginous joint, or symphysis, uniting the manubrium and body of the sternum, and it commonly ossifies with age.

86. Which muscle layer of the intercostal space assists in expiration by depressing the ribs?

A — Internal intercostal muscle (interosseous part)

The interosseous part of the internal intercostal muscle depresses the ribs, assisting expiration, in contrast to the external intercostal muscle which elevates the ribs during inspiration.

87. Which structure is described as the innermost of the three intercostal muscle layers, lying deep to the neurovascular bundle?

C — Innermost intercostal muscle

The innermost intercostal muscle forms the deepest of the three intercostal layers, lying internal to the neurovascular bundle, and is sometimes incomplete or discontinuous across the space.

88. Which structure lies in the plane between the internal intercostal and innermost intercostal muscles?

B — The intercostal neurovascular bundle

The intercostal neurovascular bundle, comprising the vein, artery, and nerve, runs in the plane between the internal intercostal muscle and the innermost intercostal muscle.

89. Which structure of the thoracic wall is a thin layer of connective tissue lining the internal surface, between the innermost intercostal muscles and the parietal pleura?

D — Endothoracic fascia

The endothoracic fascia is a thin connective tissue layer lining the internal thoracic wall, lying between the innermost intercostal muscles and the parietal pleura.

07 ANTERIOR ABDOMINAL WALL & HERNIA

LECTURE 9

90. Which muscle layer is the innermost of the three flat muscles of the anterolateral abdominal wall?

C — Transversus abdominis

Transversus abdominis is the innermost of the three flat muscles, lying deep to internal oblique, with its fibres running transversely around the abdomen.

91. Which muscle layer is the outermost of the three flat muscles of the anterolateral abdominal wall?

C — External oblique

External oblique is the outermost and largest of the three flat muscles, with fibres running downward and medially, like putting hands in trouser pockets.

92. Which layer of the abdominal wall lines the inner aspect of the cavity, deep to transversus abdominis?

E — Transversalis fascia

The transversalis fascia lines the inner aspect of the abdominal wall deep to transversus abdominis, continuing as the internal spermatic fascia and forming the anterior wall of the femoral sheath.

93. The superficial fascia of the lower anterior abdominal wall splits into two layers. What is the deeper, membranous layer called?

B — Scarpa fascia

The superficial fascia below the umbilicus splits into a superficial fatty layer, Camper fascia, and a deeper membranous layer, Scarpa fascia, which continues into the perineum as Colles fascia.

94. Scarpa fascia fuses to the deep fascia of the thigh at which structure, preventing extravasated fluid from tracking into the thigh?

C — Just below the inguinal ligament

Scarpa fascia fuses firmly with the deep fascia of the thigh (fascia lata) just below the inguinal ligament, which is why extravasated urine or fluid tracks up the abdominal wall instead of into the thigh.

95. Which two muscles form the posterior abdominal wall medially, alongside quadratus lumborum more laterally?

C — Psoas major and iliacus

Psoas major and iliacus form the posterior abdominal wall medially, with quadratus lumborum forming it more laterally, all lying against the vertebral column and pelvis.

96. The abdominal cavity is bounded superiorly by which structure?

A — Diaphragm

The abdominal cavity is bounded superiorly by the diaphragm and inferiorly by the pelvic inlet, with the iliac crest serving as a useful lateral surface landmark along the way.

97. Which nerve, the anterior ramus of the twelfth thoracic spinal nerve, runs along the inferior border of the twelfth rib?

E — Subcostal nerve

The subcostal nerve is the anterior ramus of the twelfth thoracic spinal nerve, running along the inferior border of the twelfth rib before continuing onto the abdominal wall.

98. Which nerve is the superior terminal branch of the first lumbar anterior ramus, supplying skin over the iliac crest?

D — Iliohypogastric nerve

The iliohypogastric nerve is the superior terminal branch of the first lumbar anterior ramus, supplying the skin over the iliac crest and the hypogastric region.

99. Which nerve is the inferior terminal branch of the first lumbar anterior ramus, and unlike its companion nerve, actually traverses the inguinal canal?

B — Ilioinguinal nerve

The ilioinguinal nerve is the inferior terminal branch of the first lumbar anterior ramus. Unlike the iliohypogastric nerve, it traverses the inguinal canal and emerges through the superficial ring to supply the genital skin.

100. The rectus sheath encloses which muscle along most of its length?

B — Rectus abdominis

The rectus sheath, formed by the aponeuroses of the three flat muscles, encloses rectus abdominis (and pyramidalis, where present) along most of the length of the anterior abdominal wall.

101. The linea alba runs in the midline from the xiphoid process to which structure below?

E — Pubic symphysis

The linea alba is the midline fibrous raphe formed by the fusion of the aponeuroses of the three flat muscles from both sides, extending from the xiphoid process to the pubic symphysis.

102. Which structure marks the boundary above which the posterior rectus sheath is present, and below which it is absent?

E — Arcuate line

The arcuate line marks the point below which all three flat-muscle aponeuroses pass in front of rectus abdominis, so no posterior rectus sheath exists below this line, only transversalis fascia.

08 MUSCLE PHYSIOLOGY

LECTURES 10-12

103. The sliding filament mechanism describes contraction as resulting from which process?

D — Thick and thin filaments sliding past each other

The sliding filament mechanism explains that muscle shortening occurs because the thick and thin filaments slide past one another, increasing their overlap, without either filament actually changing length.

104. During muscle shortening according to the sliding filament mechanism, what happens to the length of the individual filaments themselves?

C — Neither filament changes length

Neither the thick nor thin filaments change their own length during contraction; the sarcomere shortens purely because the two sets of filaments slide past each other, increasing their region of overlap.

105. Which zone of the sarcomere narrows as thin filaments slide further into the region between thick filaments during contraction?

D — H zone

The H zone, the central region of the A band containing only thick filaments, narrows as the thin filaments slide further inward during contraction, eventually disappearing at full shortening.

106. Which motor unit type is recruited first when a muscle contraction begins, according to the size principle?

D — Small, slow-twitch motor units

According to the size principle, the smallest motor neurons, innervating slow-twitch motor units, are recruited first as a contraction begins, with larger fast-twitch units recruited only as more force is needed.

107. Slow-twitch motor units are characteristically small and control which type of movement?

B — Fine, precise movements

Slow-twitch motor units, being small with a low innervation ratio, are suited to fine, precise control, whereas large fast-twitch units with a high innervation ratio produce powerful but less precisely graded movements.

108. Which structure of the neuromuscular junction contains vesicles of acetylcholine, forming the presynaptic portion?

D — Axon terminal

The presynaptic portion of the neuromuscular junction is the axon terminal, containing synaptic vesicles filled with acetylcholine ready for release.

109. Which structure of the neuromuscular junction bears the nicotinic acetylcholine receptors, forming the postsynaptic portion?

D — Postjunctional folds

The postsynaptic portion of the neuromuscular junction consists of the postjunctional folds of the muscle fibre membrane, which bear the nicotinic acetylcholine receptors.

110. What is the narrow space called that separates the nerve terminal from the muscle fibre at the neuromuscular junction?

A — Synaptic cleft

The synaptic cleft is the narrow space separating the presynaptic nerve terminal from the postsynaptic muscle fibre membrane at the neuromuscular junction.

111. A greater frequency of action potentials reaching the motor axon terminal produces which effect at the neuromuscular junction?

C — A higher concentration of acetylcholine in the synaptic cleft

As the frequency of motor neuron action potentials increases, acetylcholine reaches a higher concentration in the neuromuscular junction, since more vesicles are released per unit time.

112. Myasthenia gravis exists in two major clinical forms. Which form involves weakness of only the extraocular muscles?

D — Ocular myasthenia gravis

The two major forms of myasthenia gravis are weakness confined to the extraocular muscles (ocular myasthenia) and generalized weakness affecting all skeletal muscles.

113. In myasthenia gravis, anti-acetylcholine receptor antibodies cause degeneration of which structure?

E — Postjunctional folds

In myasthenia gravis, spontaneous production of anti-acetylcholine receptor antibodies results in progressive loss of muscle receptors and degeneration of the postjunctional folds of the motor end plate.

114. Which structure senses depolarisation of the transverse tubule and mechanically opens the calcium release channel of the sarcoplasmic reticulum?

E — Dihydropyridine receptor

The dihydropyridine receptor of the transverse tubule membrane senses depolarisation and is mechanically linked to the ryanodine receptor of the sarcoplasmic reticulum, directly triggering calcium release.

115. Which receptor of the sarcoplasmic reticulum is directly opened by the dihydropyridine receptor to release calcium?

E — Ryanodine receptor

The ryanodine receptor of the sarcoplasmic reticulum is the calcium release channel, mechanically coupled to and opened by the dihydropyridine receptor of the transverse tubule.

116. Once released into the cytoplasm, calcium is taken back up by which structure to allow relaxation?

D — Sarcoplasmic reticulum

Calcium is reaccumulated by the sarcoplasmic reticulum via an active calcium pump, lowering cytoplasmic calcium and allowing the muscle fibre to relax.

117. Which molecule directly binds to the myosin head to allow it to detach from actin during the cross-bridge cycle?

A — Adenosine triphosphate

A fresh molecule of adenosine triphosphate must bind to the myosin head for it to detach from actin, which is why the absence of ATP after death leaves the heads locked on, producing rigor mortis.

118. Hydrolysis of adenosine triphosphate on the myosin head provides the energy for which step of the cross-bridge cycle?

B — Re-cocking the myosin head for the next power stroke

Hydrolysis of adenosine triphosphate on the myosin head provides the energy to re-cock the head into a high-energy position, ready for the next power stroke once it rebinds actin.

119. Which ion, once bound to troponin C, moves tropomyosin away from the myosin-binding sites on actin?

B — Calcium

Calcium binding to troponin C causes a conformational change that shifts tropomyosin away from the myosin-binding sites on the actin filament, allowing cross-bridges to form.

120. Which part of the sarcomere represents the full length of the thick filaments and stays constant in width during contraction?

A — A band

The A band represents the full length of the thick filaments and stays constant in width during contraction, unlike the I band and H zone, which both narrow as the filaments increase their overlap.

Answer Grid

For fast marking

No.	Ans	No.	Ans	No.	Ans	No.	Ans	No.	Ans
1	D	25	D	49	E	73	E	97	E
2	C	26	E	50	C	74	A	98	D
3	C	27	C	51	C	75	C	99	B
4	E	28	A	52	A	76	E	100	B
5	C	29	A	53	B	77	A	101	E
6	E	30	B	54	D	78	B	102	E
7	D	31	A	55	B	79	C	103	D
8	B	32	E	56	A	80	C	104	C
9	B	33	E	57	D	81	C	105	D
10	B	34	A	58	B	82	D	106	D
11	A	35	E	59	E	83	A	107	B
12	D	36	D	60	B	84	D	108	D
13	E	37	A	61	A	85	B	109	D
14	C	38	C	62	D	86	A	110	A
15	A	39	D	63	C	87	C	111	C
16	A	40	E	64	C	88	B	112	D
17	B	41	B	65	A	89	D	113	E
18	C	42	A	66	E	90	C	114	E
19	A	43	A	67	C	91	C	115	E
20	B	44	D	68	D	92	E	116	D
21	D	45	E	69	C	93	B	117	A
22	E	46	E	70	B	94	C	118	B
23	B	47	B	71	A	95	C	119	B
24	D	48	B	72	E	96	A	120	A

*Distribution of correct answers — **A** 24 · **B** 24 · **C** 24 · **D** 24 · **E** 24 · total **120**. Exactly even, so the position of an answer tells you nothing.*